AI-POWERED GRAIN CLASSIFICATION

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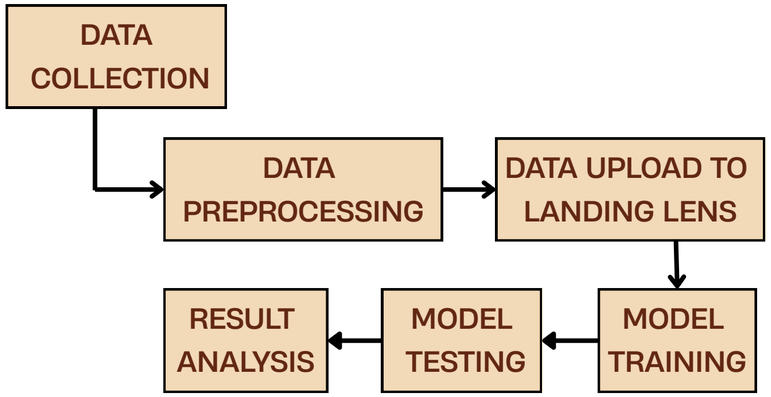
1. ABSTRACT

This project presents an AI-driven grain classification system leveraging LandingLens to accurately identify wheat, rice, barley, rajma, and chana from images. By analyzing distinctive grain features, the system automates sorting processes, providing a fast, reliable, and scalable solution for agriculture and food processing industries.

2. INTRODUCTION

Grain classification is essential for quality control in agriculture and food processing. Manual sorting is slow and error-prone, while AI provides faster, more accurate solutions. This project leverages LandingLens, a low-code visual AI platform, to classify wheat, rice, barley, rajma, and chana from images, delivering efficient and reliable grain identification with minimal technical complexity.

3. METHODOLOGY



4. INPUT / OUTPUT

Test Image	Actual	Predicted	Result
	Wheat	Wheat	✓
	Rajma	Rajma	✓
	Rice	Rice	✓
	Chana	Chana	✓
	Barley	Barley	✓

5. RESULT

The system achieved high classification accuracy across multiple grain types, with minimal misclassification between similar grains like wheat and barley. These results demonstrate the model's effectiveness and reliability for real-world grain identification tasks.

6. CONCLUSION

The system supports seamless deployment in agricultural and food processing environments, enabling automated grain sorting, quality control, and seed verification. Its real-time operation allows efficient, non-intrusive monitoring of grain batches, helping reduce manual labor, minimize errors, and optimize workflow across the grain supply chain.

7. STEPS TO USE THE APPLICATION

Step-1. Click / Save a picture of a grain sample as shown below.

Step-2. Scan the QR code.

Step-3. Choose Any Photo – Select a sample image from the gallery. (or Take a live photo)

Step-4. Click "Use Photo" and Get instant grain classification result.

8. REFERENCES

1. Zhang, Y., & Wang, S. (2019). Grain classification using convolutional neural networks. Computers and Electronics in Agriculture, 162, 431–439.
2. Singh, A., & Kaur, R. (2020). Machine learning-based grain quality assessment system. IEEE Access, 8, 145698–145707.
3. Landing AI - Landing Lens Support Documentation- Officical Website: support.landing.ai